Kubernetes-05

Todays topics coverage :

Namespace

- Logical separation inside one Kubernetes cluster
- Used to create environments (dev, staging, prod)
- Same resource names allowed in different namespaces
- Provides isolation between teams/projects
- Helps apply policies and quotas per team
- Default namespace exists if not specified

Requests & Limits

Requests

- Minimum CPU/RAM guaranteed to container
- Scheduler uses this to place pod on node
- Pod will always get at least this much resource

Limits

- Maximum CPU/RAM container can use
- Prevents container from using entire node

When exceeded

- CPU > limit → throttled (slow)
- Memory > limit → pod killed (OOMKilled)

Requests = reservation
Limits = restriction

ResourceQuota

- Applied at **namespace level**
- Controls total resource usage of a team
- Restricts number of objects (pods, services, PVC, etc.)
- Prevents one team from consuming full cluster
- If quota exceeded → resource creation fails
- Used by admins in multi-team clusters

LimitRange

- Applied at **namespace level**
- Sets rules for each container/pod inside namespace
- Defines minimum resource allowed
- Defines maximum resource allowed
- Assigns default resources if developer doesn't specify
- Prevents creating very small or very large pods
- Works together with ResourceQuota

What it can control

- min CPU / memory
- max CPU / memory
- default CPU / memory
- default requests
- limit-to-request ratio

Why needed

- Developers may forget to add resources

- Some may request too much resources
- Keeps cluster stable and balanced

Tasks:

1. Create a namespace dev-environment and apply a resource-based quota that restricts the number of pods to 3 and services to 2.
2. Create a pod in the prod-environment namespace with 0.2 CPU and 200Mi memory requests, and 0.5 CPU and 500Mi memory limits.
3. In the staging-environment namespace, set a LimitRange that assigns default CPU and memory limits (300m CPU, 600Mi memory) and applies a minimum and maximum CPU.
4. Create a pod and a NodePort service in the default namespace, then create another pod in the test namespace and communicate between them using Service DNS.
5. Apply a LimitRange with a max limit/request ratio of 2 for memory in the performance-environment namespace, and test by creating a pod with mismatched resource requests and limits.

TASK 1 : Create a namespace dev-environment and apply a resource-based quota that restricts the number of pods to 3 and services to 2.

Create a namespace `dev-environment` and apply ResourceQuota

- pods = 3
- services = 2

A **namespace** is like a separate workspace inside your Kubernetes cluster.
It allows you to **organize** and **isolate** resources (like Pods, Services, Deployments, etc.) so that different teams or environments (e.g., dev, test, prod) don't interfere with each other.

So here, we're creating a namespace named **dev-environment** to hold all our development-related resources.

Step 1 — Create namespace

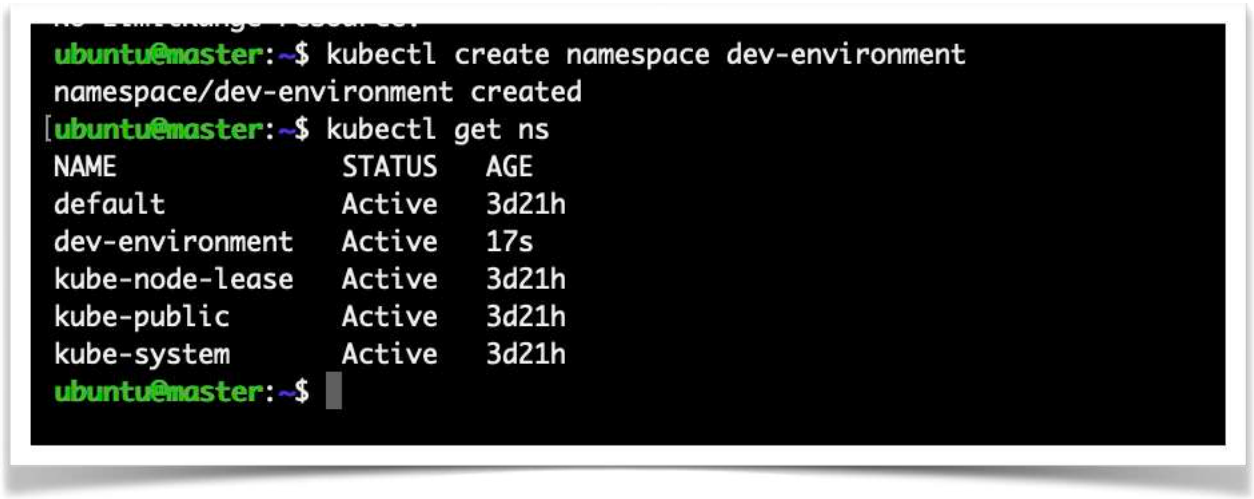
Run:

```
kubectl create namespace dev-environment
```

Check:

```
kubectl get ns
```

You should see `dev-environment`



Step 2 — Create quota YAML

Create file:

```
vi dev-quota.yml
```

```
ubuntu@master:~$ vi dev-quota.yml
ubuntu@master:~$ cat dev-quota.yml
apiVersion: v1
kind: ResourceQuota
metadata:
  name: dev-quota
  namespace: dev-environment
spec:
  hard:
    pods: "3"
    services: "2"

ubuntu@master:~$
```

Step 3 — Apply quota

`kubectl apply -f dev-quota.yml`
Verify:

`kubectl describe quota dev-quota -n dev-environment`

```
ubuntu@master:~$ kubectl apply -f dev-quota.yml
resourcequota/dev-quota created
ubuntu@master:~$ kubectl describe quota dev-quota -n dev-environment
Name:       dev-quota
Namespace:  dev-environment
Resource    Used  Hard
--------    ----  ----
pods        0     3
services    0     2
ubuntu@master:~$
```

Step 4 — Test

Create pods:

```
kubectl run pod1 --image=nginx -n dev-environment
kubectl run pod2 --image=nginx -n dev-environment
kubectl run pod3 --image=nginx -n dev-environment
kubectl run pod4 --image=nginx -n dev-environment
```

4th pod must FAIL

—-> I have used three pods now lets test with 4th pod will it create or not

Now we can see when you try to create a **fourth Pod**, you'll get an error like this:

Kubernetes is enforcing:

Team cannot exceed allocated resources

Even if cluster has free CPU/RAM → still blocked.

TASK-2: Create a pod in the prod-environment namespace with 0.2 CPU and 200Mi memory requests, and 0.5 CPU and 500Mi memory limits.

Create a **pod in `prod-environment` namespace** with:

- Request → 0.2 CPU & 200Mi memory
- Limit → 0.5 CPU & 500Mi memory

Here you will understand **how Kubernetes guarantees and restricts resources**.

STEP 1 — Create Namespace

```
kubectl create namespace prod-environment
```

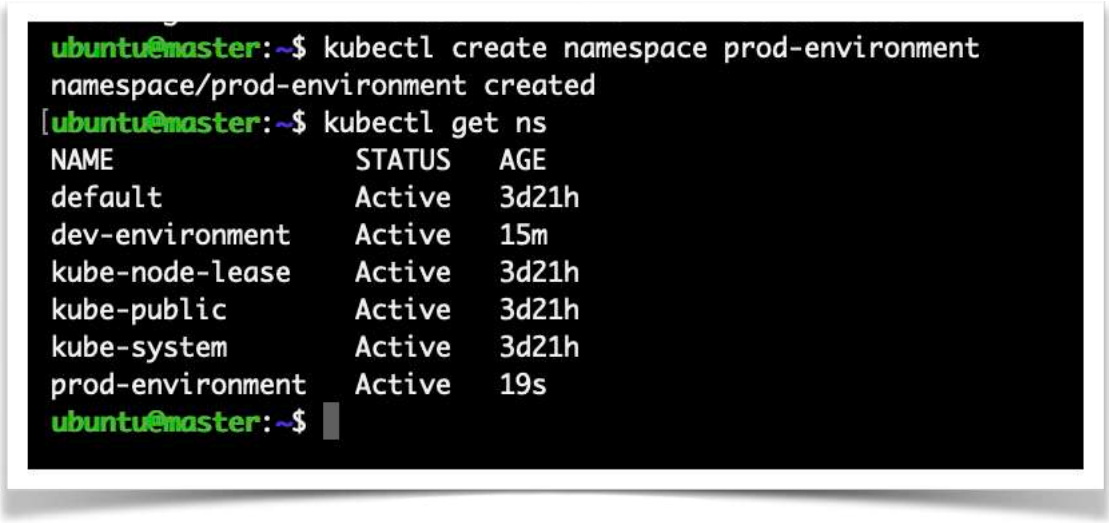
Why?

We separate real application (production) from dev testing.

```
Cluster
  ├── dev-environment
  └── prod-environment  ← real users app
```

Check:

```
kubectl get ns
```



STEP 2 — Understand Requests vs Limits (before writing YAML)

Think like this:

Resource	Meaning
Request	minimum guaranteed
Limit	maximum allowed

So we tell Kubernetes scheduler:

"My app needs at least 0.2 CPU to run properly
but never allow it to cross 0.5 CPU"

Important behavior

Condition	Result
Node doesn't have request CPU	pod NOT scheduled
CPU crosses limit	throttled (slow)
Memory crosses limit	pod killed (OOMKilled)

STEP 3 — Create Pod YAML

Open file:

```
vi prod-pod.yml
```

```
prod-environment   Active   19s
ubuntu@master:~$ vi prod-pod.yml
ubuntu@master:~$ cat prod-pod.yml
apiVersion: v1
kind: Pod
metadata:
  name: prod-pod
  namespace: prod-environment
spec:
  containers:
  - name: nginx
    image: nginx
    resources:
      requests:
        cpu: "200m"
        memory: "200Mi"
      limits:
        cpu: "500m"
        memory: "500Mi"

ubuntu@master:~$
```

Understand YAML

Field	Meaning
requests	guaranteed resources
limits	maximum usage
200m cpu	0.2 core
500m cpu	half CPU core

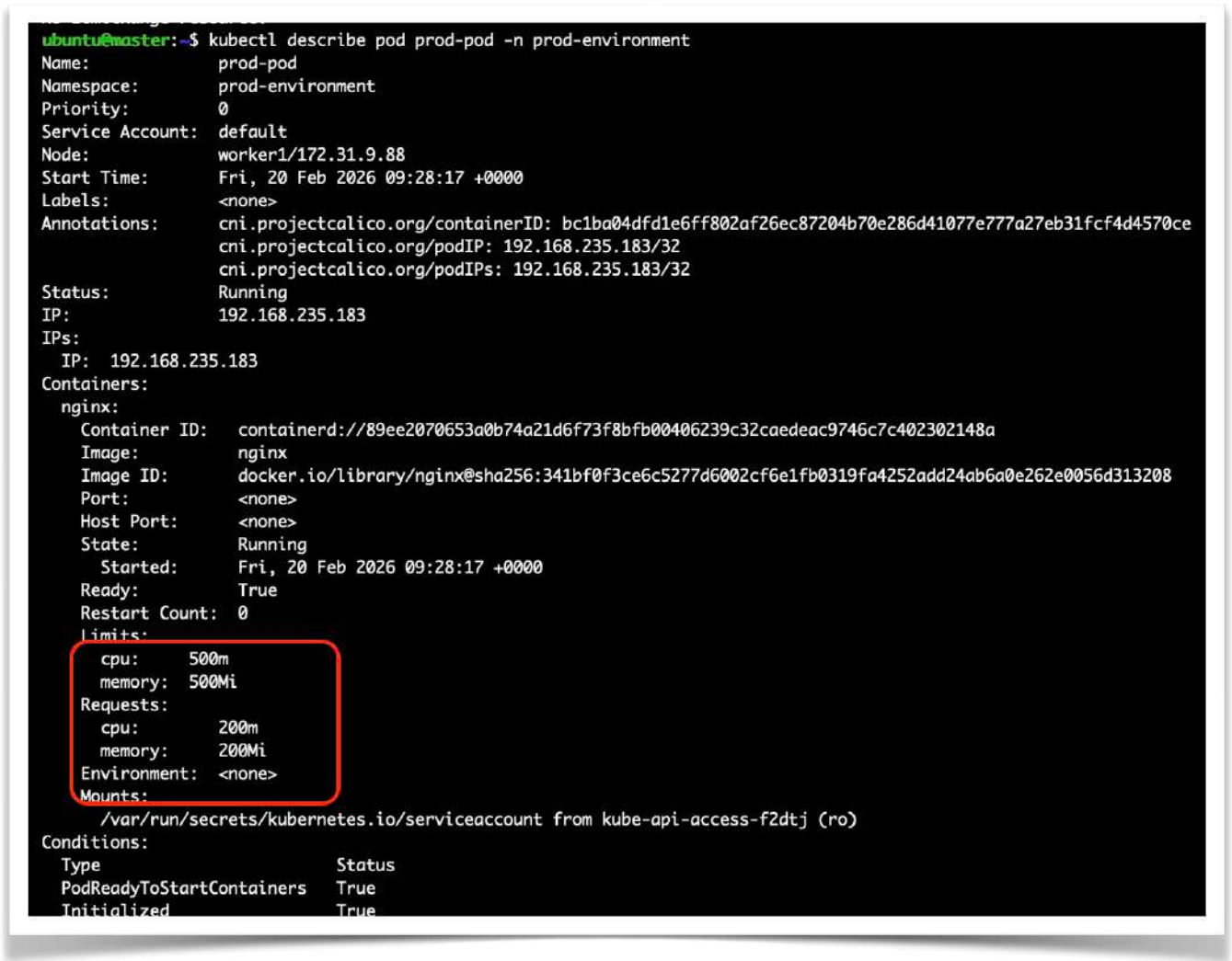
STEP 4 — Apply Pod

```
kubectl apply -f prod-pod.yml
```

STEP 5 — Verify

```
kubectl describe pod prod-pod -n prod-environment
```

```
ubuntu@master:~$ kubectl apply -f prod-pod.yml
pod/prod-pod created
```



What we proved

Scheduler now reserves CPU & RAM before running pod.

So Kubernetes now knows:

“This application needs guaranteed performance”

This is **mandatory in production clusters**.

TASK-3 — In the staging-environment namespace, set a LimitRange that assigns default CPU and memory limits (300m CPU, 600Mi memory) and applies a minimum and maximum CPU.

LimitRange

Namespace: **staging-environment**

We will create rules:

- Default CPU = 300m
- Default Memory = 600Mi
- Min CPU = 100m
- Max CPU = 800m

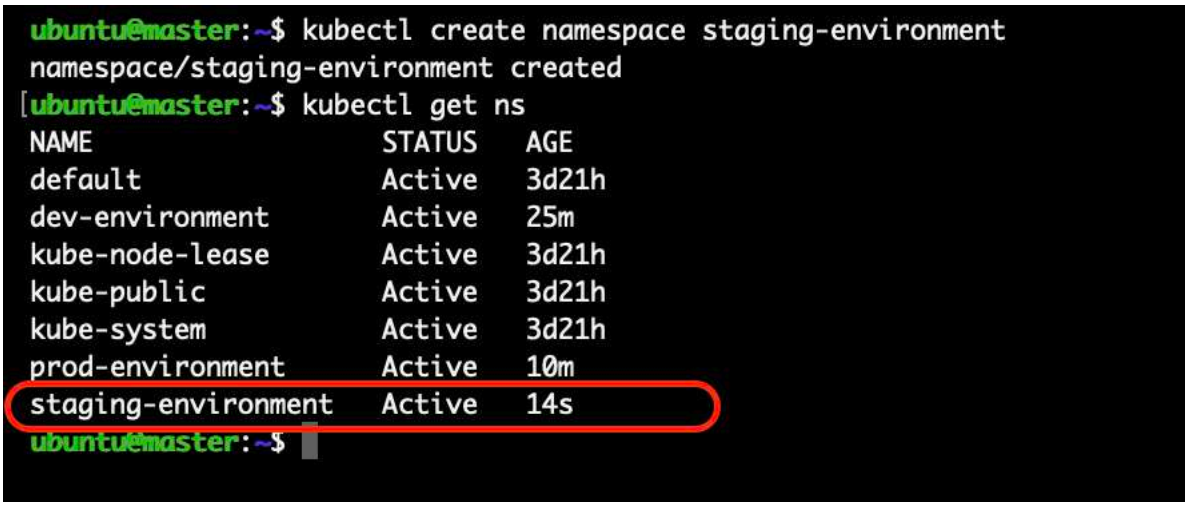
Meaning:
Even if developer forgets resources → Kubernetes auto assigns
And nobody can request too small or too large CPU

STEP 1 — Create namespace

```
kubectl create namespace staging-environment
```

Check:

```
kubectl get ns
```



STEP 2 — Why LimitRange?

Problem in companies:

Developer mistake	Result
No resources defined	Pod eats all node memory
Too small CPU	App crashes
Too big CPU	Other apps starve

So admin creates **rules per container** → LimitRange

ResourceQuota controls TOTAL
LimitRange controls EACH POD

STEP 3 — Create LimitRange YAML

Open file:

```
vi staging-limit.yml
```

Understand fields

Field	Meaning
default	auto assign limit if not provided
defaultRequest	auto assign request
min	cannot go below
max	cannot exceed

STEP 4 — Apply rule

```
kubectl apply -f staging-limit.yml
```
Verify:

```
kubectl describe limitrange -n staging-environment
```

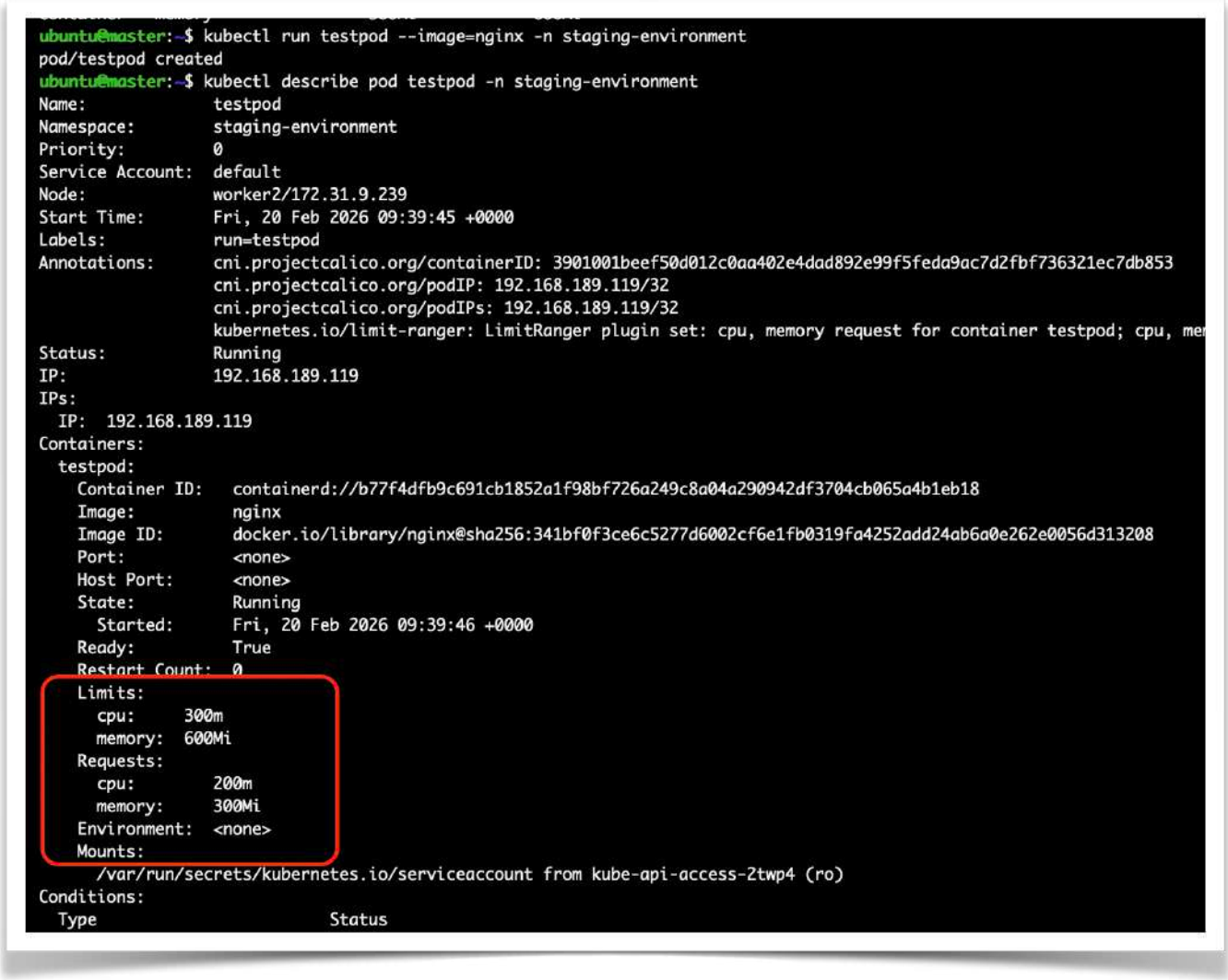
STEP 5 — TEST (important)

Now create pod WITHOUT resources:

```
kubectl run testpod --image=nginx -n staging-environment
```

Check assigned resources:

```
kubectl describe pod testpod -n staging-environment
```
You will see automatically:



Kubernetes added values automatically ✔

What we proved

LimitRange forces developers to follow company policy.

Even if YAML empty → cluster safe.

TASK-4 :Create a pod and a NodePort service in the default namespace, then create another pod in the test namespace and communicate between them using Service DNS.

Communication using Service DNS
(This is VERY real-time — apps talk to each other like frontend → backend → database)

We will:

1. Create **web server pod + NodePort service (default namespace)**
2. Create **client pod in test namespace**
3. Access it using **Kubernetes internal DNS**

STEP 1 — Create nginx pod YAML (default namespace)

Create file:

```
vi web-pod.yml
```

```
ubuntu@master:~$ vi web-pod.yml
ubuntu@master:~$ cat web-pod.yml
apiVersion: v1
kind: Pod
metadata:
  name: web-pod
  labels:
    app: web
spec:
  containers:
  - name: nginx
    image: nginx
    ports:
    - containerPort: 80

ubuntu@master:~$
```

Apply:

```
kubectl apply -f web-pod.yml
kubectl get pods
```

```
ubuntu@master:~$ kubectl apply -f web-pod.yml
pod/web-pod created
ubuntu@master:~$ kubectl get pods
NAME      READY   STATUS    RESTARTS   AGE
web-pod   1/1     Running   0          8s
ubuntu@master:~$
```

STEP 2 — Create service YAML

Create file:

```
vi web-service.yml
```

```
ubuntu@master:~$ vi web-service.yml
ubuntu@master:~$ cat web-service.yml
apiVersion: v1
kind: Service
metadata:
  name: web-service
spec:
  type: NodePort
  selector:
    app: web
  ports:
  - port: 80
    targetPort: 80

ubuntu@master:~$
```

Apply:

```
kubectl apply -f web-service.yml
kubectl get svc
```

```
ubuntu@master:~$ kubectl apply -f web-service.yml
service/web-service created
ubuntu@master:~$ kubectl get svc
NAME          TYPE        CLUSTER-IP     EXTERNAL-IP   PORT(S)        AGE
kubernetes    ClusterIP   10.96.0.1      <none>        443/TCP        22h
web-service   NodePort    10.98.106.72   <none>        80:32584/TCP   14s
ubuntu@master:~$
```

What happened internally

Service checks labels:

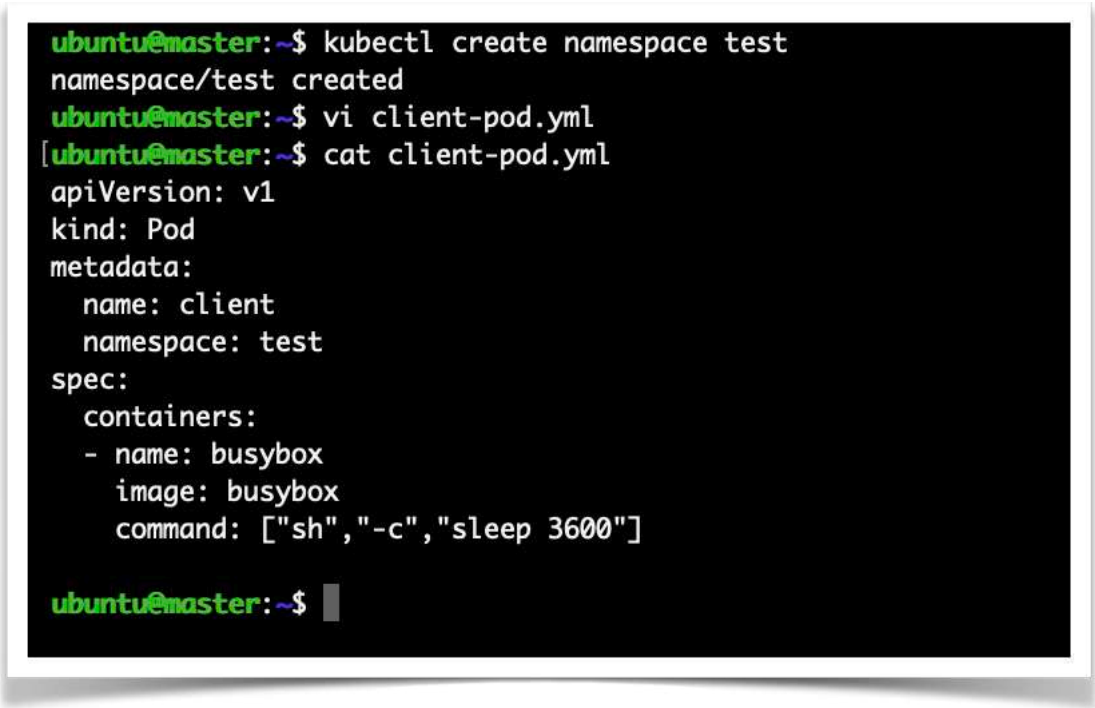
```
Service selector: app=web
        ↓
Pod label: app=web
        ↓
Traffic connected
```

STEP 3 — Create test namespace

```
kubectl create namespace test
```

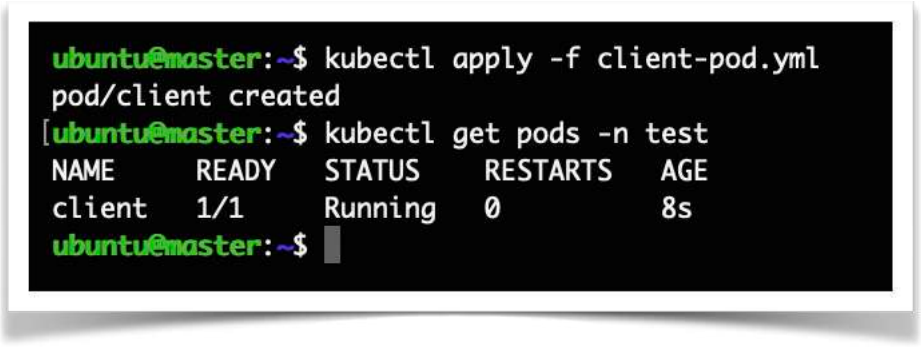
STEP 4 — Create client pod YAML

```
vi client-pod.yml
```



Apply:

```
kubectl apply -f client-pod.yml
kubectl get pods -n test
```



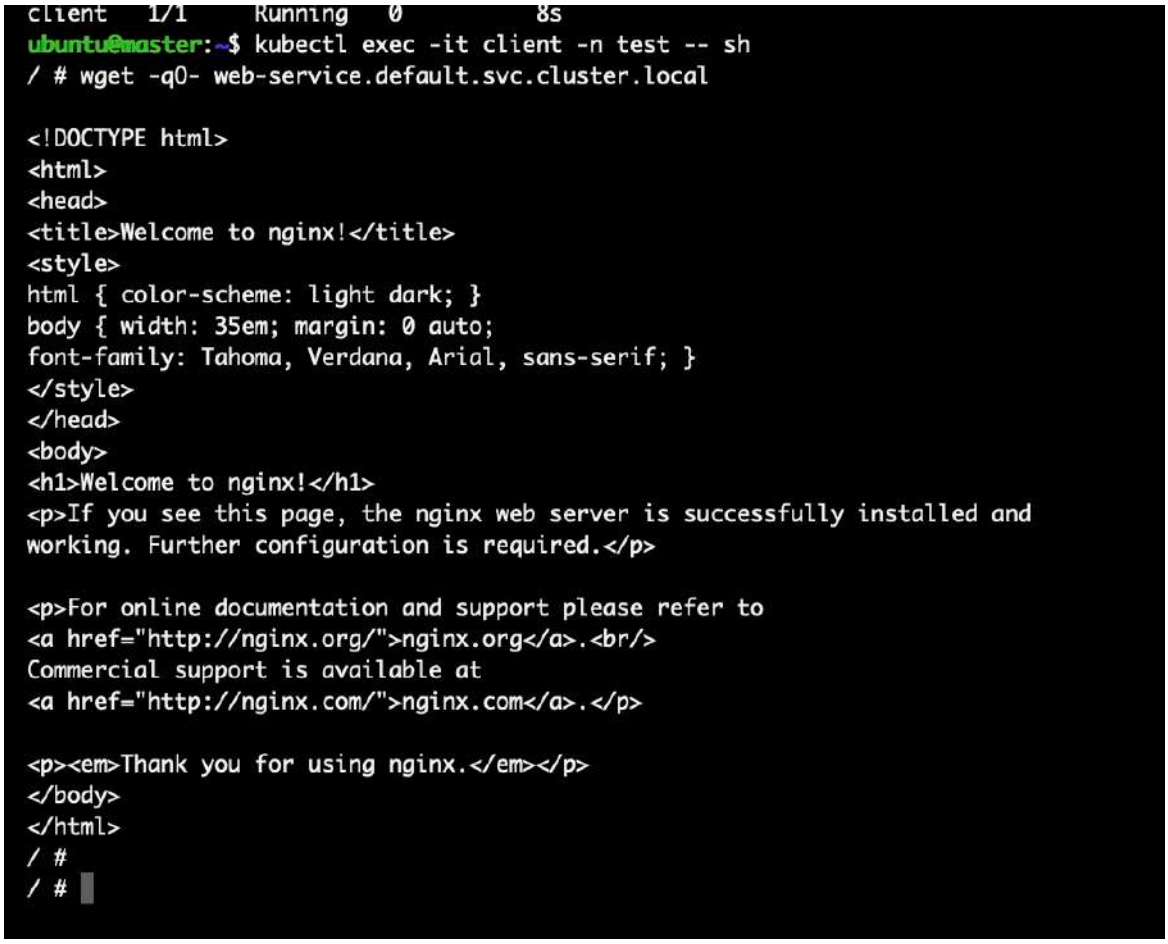
STEP 5 — Enter client container

```
kubectl exec -it client -n test -- sh
```

STEP 6 — DNS communication test

Inside container:

```
wget -qO- web-service.default.svc.cluster.local
```



Expected output

You will see nginx HTML page:

```
Welcome to nginx!
```

TASK-5 — Apply a LimitRange with a max limit/request ratio of 2 for memory in the performance-environment namespace, and test by creating a pod with mismatched resource requests and limits.

LimitRange *limit-to-request ratio*

Namespace: **performance-environment**

We will enforce rule:

A container's **memory limit cannot be more than 2× its request**

First understand the concept

Example:

Request	Limit	Allowed?
200Mi	300Mi	✔ OK
200Mi	400Mi	✔ OK
200Mi	500Mi	❌ NOT OK

Because:

```
limit / request = ratio
500 / 200 = 2.5 > allowed 2
```

So Kubernetes blocks greedy containers.

This prevents developers from doing:

reserve very small memory but try to consume huge memory

STEP 1 — Create namespace

```
kubectl create namespace performance-environment
```

Check:

```
kubectl get ns
```

```
ubuntu@master:~$ kubectl create namespace performance-environment
namespace/performance-environment created
ubuntu@master:~$ kubectl get ns
NAME                      STATUS   AGE
default                   Active   3d22h
dev-environment           Active   53m
kube-node-lease           Active   3d22h
kube-public               Active   3d22h
kube-system               Active   3d22h
performance-environment   Active   7s
prod-environment          Active   38m
staging-environment       Active   28m
test                      Active   13m
ubuntu@master:~$
```

STEP 2 — Create LimitRange rule

Create file:

```
vi ratio-limit.yml
```

```
ubuntu@master:~$ vi ratio-limit.yml
ubuntu@master:~$ cat ratio-limit.yml
apiVersion: v1
kind: LimitRange
metadata:
  name: memory-ratio-limit
  namespace: performance-environment
spec:
  limits:
  - type: Container
    maxLimitRequestRatio:
      memory: "2"

ubuntu@master:~$
```

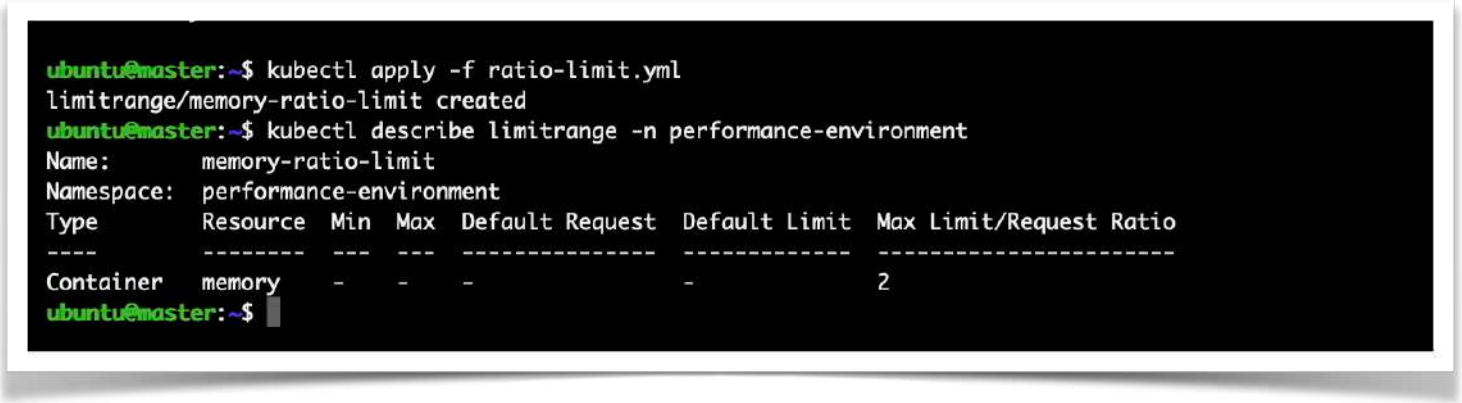
Meaning

Kubernetes now enforces:

```
memory limit ≤ 2 × memory request
```

Apply:

```
kubectl apply -f ratio-limit.yml
kubectl describe limitrange -n performance-environment
```



STEP 3 — Create BAD pod (should fail)

Create file:

```
vi bad-pod.yml
```

```
[ubuntu@master:~$ vi bad-pod.yml
[ubuntu@master:~$ cat bad-pod.yml
apiVersion: v1
kind: Pod
metadata:
  name: badpod
  namespace: performance-environment
spec:
  containers:
  - name: nginx
    image: nginx
    resources:
      requests:
        memory: "100Mi"
      limits:
        memory: "500Mi"

ubuntu@master:~$
```

Why this fails

```
limit = 500Mi
request = 100Mi

ratio = 5
allowed = 2
```

Apply:

```
kubectl apply -f bad-pod.yml
```

```
ubuntu@master:~$ kubectl apply -f bad-pod.yml
Error from server (Forbidden): error when creating "bad-pod.yml": pods "badpod" is forbidden: memory max limit to request ratio per Container is 2, but provided ratio is 5.000000
ubuntu@master:~$
```

Expected error

You will get something like:

```
Error from server (Forbidden):
maximum memory limit to request ratio per Container is 2
```

What we proved

LimitRange can also control **fairness**, not just min/max/default.

Prevents:

- cheating resource allocation
- node memory pressure
- noisy neighbor problem

Kubernetes Lab — Topics Covered

Namespace (environment separation)

- Created multiple namespaces (dev, prod, staging, test, performance)
- Learned same resource name can exist in different namespaces
- Used `-n <namespace>` to isolate applications

Purpose: separate teams & environments

ResourceQuota (team usage control)

- Limited namespace to **3 pods & 2 services**
- Tried creating 4th pod → blocked
- Understood quota controls **total usage**

Prevents one team from consuming whole cluster

Requests & Limits (container resource control)

- Created production pod with CPU & memory guarantees
- Understood scheduler reserves request resources
- Learned behavior:
  - CPU exceed → throttled
  - Memory exceed → OOMKilled

Used in real production apps

LimitRange (rules per container)

- Applied default CPU & memory automatically
- Set min & max CPU
- Pod created without resources → auto assigned values

Forces developers to follow company policy

Service & DNS Communication

- Created pod + service
- Accessed from another namespace
- Used internal DNS:

```
service.namespace.svc.cluster.local
```

- Tested using wget/curl inside pod

Learned: Pods never communicate using pod IP

LimitRange Ratio (advanced control)

- Enforced memory limit/request ratio = 2
- Created invalid pod → Kubernetes rejected

Prevents resource abuse (“reserve less, consume more”)

Big Concepts You Learned

Level	What it controls
Namespace	environment separation
ResourceQuota	total team usage
Requests/Limits	container resources
LimitRange	rules per container
Service DNS	pod communication
Ratio limit	fairness & stability